SHAZIN

Final-Year Computer Engineering Student | Embedded Systems, Robotics & Software

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PROFILE

Final-year Computer Engineering student with a 3.63/4.00 CGPA and two Vice-Chancellor's List awards. My work includes firmware for satellite-connected sensors, a robot-fleet simulator, and contributions to OrcaSlicer. I'm interested in low-power sensors, robot coordination, and simulation-based optimisation. I also work on Linux projects outside my degree.

EDUCATION & ACADEMIC HONOURS

Asia Pacific University of Technology & Innovation (APU) — **2022 - Expected 2026**

Bachelor of Engineering in Computer Engineering | Final semester

- Current CGPA: 3.63/4.00 through Year 4, Semester 1.
- Vice-Chancellor's List for Excellent Academic Achievement: Academic Year 2023/2024 and Academic Year 2024/2025.

INDUSTRY EXPERIENCE

Averis - Engineering Intern (Embedded Systems & IoT) — **Jun - Oct 2025**

Low-power sensor networks, satellite connectivity, prototyping and technical documentation

- Developed C++ firmware for custom STM32U0 ground nodes that read water-level sensors over Modbus/RS-485, pack telemetry and transmit it through an Iridium Short Burst Data satellite modem.
- Implemented RTC-controlled shutdown and deep sleep to reduce power use. Added battery monitoring, automatic sensor detection, and FRAM/EEPROM storage to queue unsent readings for the next connection.
- Tested ground nodes in field deployments, wrote handover documentation, and guided incoming interns. Also standardised FDM printer calibration and OrcaSlicer profiles and prototyped node enclosures.

ENGINEERING PROJECTS

Agricultural Robot Fleet Discrete-Event Simulator - Final Year Project — **2026 - Present**

Simulation-based optimisation of heterogeneous oil-palm harvesting fleets

- Developing a simulation to compare fleet sizes and charging capacity for oil-palm harvesting. The model accounts for how Langur harvesters, Howler collectors, and Mandril transporters depend on one another.
- Built a SimPy model using plantation data and Delaunay cells, with battery limits, charging, and drainage-aware routing. Compared Hungarian task assignment with greedy and field-sequential methods.
- Built a FastAPI/SQLite backend and React/TypeScript dashboard to watch fleet state over WebSockets and compare run results. Repeatable parameter sweeps cover robot ratios, fleet size, and charging-station count.

Autonomous Drone for Soft Payload Deployment and Pickup — **2026**

Four-person university group project | GUI dashboard and communication protocol developer

- Built the Raspberry Pi 5 MAVLink backend, WebSocket server, and browser dashboard to monitor flight state, mission progress, payload status, and ArUco markers.
- Used MAVProxy to share telemetry over UDP and resolve MAVLink serial conflicts. Compiled Intel RealSense support for Raspberry Pi from source and adjusted telemetry rates to reduce CPU load before integration.
- Helped write and carry out autonomous-flight tests with pre-flight checks, failsafe monitoring, manual overrides, and procedures for aborts and loss of control. Integrated testing recorded a peak payload impact force of 9 N.

OPEN SOURCE & INDEPENDENT ENGINEERING

OrcaSlicer FullSpectrum / Snapmaker Orca Integration **2025 - 2026**

- Worked with ratdoux and other contributors on colour mixing and local-Z dithering in OrcaSlicer's C++ codebase. Snapmaker merged our code into its official slicer in PR #409. Upstream OrcaSlicer also credits our contribution in PR #13437.
- Used Git to collaborate on the changes and helped with integration, testing, documentation, and ongoing maintenance.

Custom Android updates for Samsung Galaxy Note 5 **2020**

- Built and maintained an official Android 10 ROM for multiple Note 5 variants using device trees, kernel, and vendor sources. Managed releases, updates, and user support. The XDA Developers release page passed 10,000 downloads.

Armbian Linux Ports for Repurposed Android Phones **2025**

- Ported Armbian to several Android phones and tested them as low-cost home servers and Raspberry Pi alternatives. Used their batteries as backup power and USB breakout hardware for GPIO/UART connections.

Lifestitchr Atlas - Web Deckbuilder **2026**

- Built and deployed a responsive deckbuilder for an original card game, with card search/filtering, deck rules, starter decks, previews, shareable URL state and automated card-atlas generation from source assets.

ADDITIONAL ACADEMIC PROJECTS

- Digital/FPGA design: implemented and tested code converters, multiplexers and demultiplexers in VHDL on an Intel Cyclone IV E FPGA using Quartus.
- Analogue design: designed and breadboard-tested BJT amplifier circuits and used LTspice for simulation and analysis.
- Embedded: built an ESP32 web-based aircraft infotainment prototype.
- Computer vision: built a YOLO-based plant-disease identification system.

TECHNICAL SKILLS

Programming: C/C++, Python, TypeScript/JavaScript, VHDL, SQL, HTML/CSS

Embedded & hardware: STM32, ESP32/Arduino, UART, I2C, RS-485/Modbus, Iridium SBD, low-power firmware, sensor integration, FPGA

Software & systems: Git/GitHub, Linux, Docker, FastAPI, React, SimPy, SQLite, REST, WebSockets/SSE, testing and documentation. Arch Linux is my daily desktop OS.

EDA, CAD & fabrication: LTspice, Intel Quartus, Cadence Virtuoso (basic), SolidWorks, Autodesk Fusion, OrcaSlicer and FDM 3D printing

ADDITIONAL TRAINING

- Completed AWS Cloud Architecting coursework as part of APU's cloud computing module.